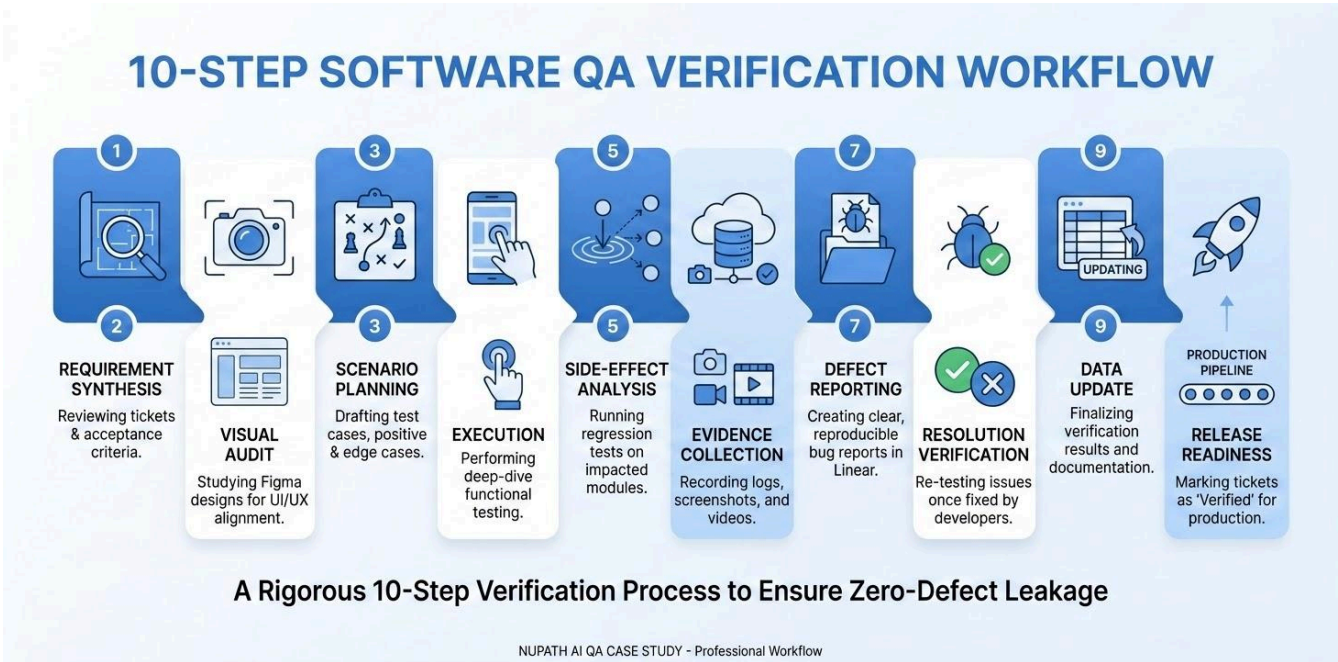


NuPath AI: QA Case Study

Elevating Product Reliability Through Comprehensive Quality Engineering

Professional Summary: Results-driven Quality Assurance Engineer specializing in mobile health ecosystems and AI-driven platforms. Proven expertise in end-to-end validation, cross-platform regression testing, and defect lifecycle management within Agile frameworks. Dedicated to ensuring seamless user experiences through meticulous visual audits and technical rigor.

Quality Assurance Excellence in Health & Wellness Tech



Prepared by: Shawon Ahmed

Role: Software Quality Assurance Engineer

Date: Aug 2, 2026 12:00 AM

1. Project Executive Summary

NuPath AI is a sophisticated mobile health and wellness platform designed to facilitate holistic well-being through AI-driven coaching and data-centric health tracking. The ecosystem encompasses dual-platform mobile applications (Android and iOS), wearable device integrations, and a robust Admin Portal for high-level analytics and management.

As a QA Engineer, I was tasked with maintaining the integrity of the user experience by validating complex AI workflows, ensuring cross-platform feature parity, and managing the defect lifecycle within an agile development environment.

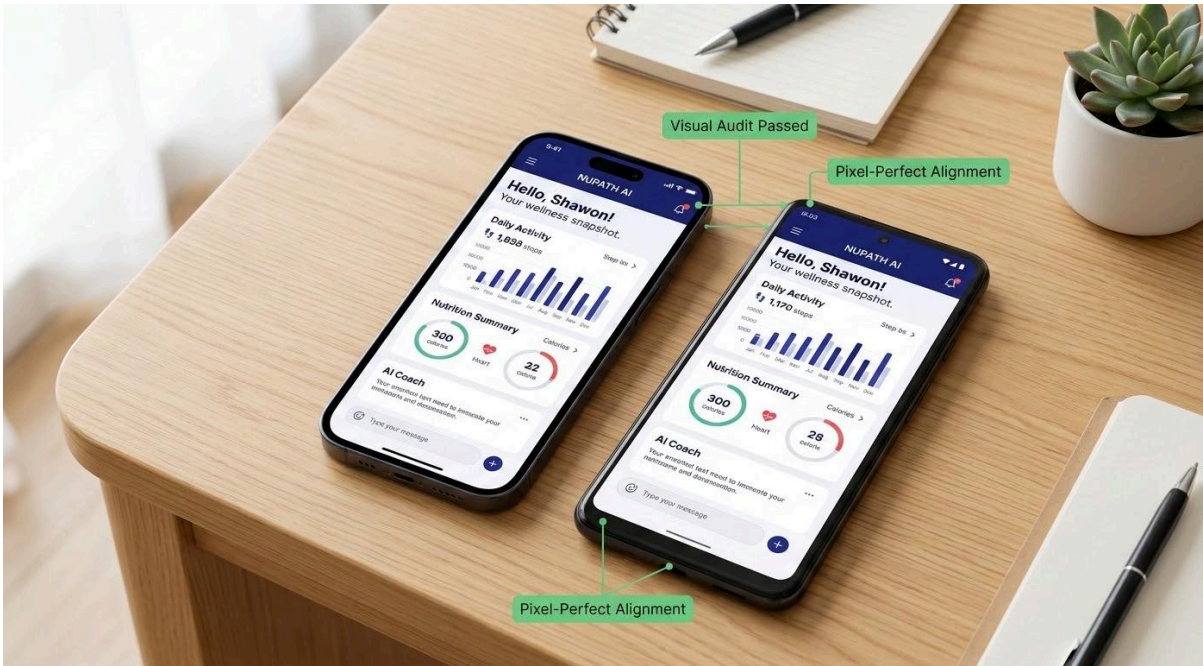
2. Strategic Testing Scope

My contribution focused on a multi-dimensional testing strategy to ensure the reliability of the application's core value propositions.

Core Ecosystem Coverage

- **Mobile Platforms:** Full-cycle validation of Android and iOS builds.
- **User Journeys:** End-to-end verification of onboarding, daily health logging, and AI coaching.
- **Integrations:** Wearable device data synchronization and API-integrated feature health.
- **Infrastructure:** Admin Portal management tools and production-ready build verification.

Feature-Specific Validation



Category	Validated Features
Authentication	Onboarding, Sign-in/Sign-up, Account Settings
Health Tracking	Nutrition Log, Activity Log, Sleep, Substance Logging
AI & Social	AI Coach (Chat/Voice), Locker Room Community, Direct Messaging
Gamification	Reward & Power Points System, Goals & Reflections
Operations	Analysis Dashboard, Notifications, Offline Mode, Admin Portal

3. Methodology & Technical Stack

I employed a structured QA workflow to maintain high standards of documentation and bug traceability.

Testing Types Executed

- **Functional & Regression:** Verifying new features and ensuring existing modules remained stable during updates.
- **UI/UX Validation:** Ensuring pixel-perfect implementation against Figma prototypes.
- **Platform Specific:** Cross-platform, Background/Resume, and Offline capability testing.
- **Boundary Analysis:** Edge case and negative testing to verify system resilience.

Technical Toolkit

- **Management:** Linear (Defect Tracking/Sprints), Google Sheets.
- **Development Tools:** Android Studio, TestFlight, Postman (API Testing).
- **Design & Communication:** Figma, Slack.
- **Documentation:** Screen Recording and Screenshot annotation tools.

SDLC Integration & Agile Alignment

My QA process was tightly woven into the Software Development Life Cycle (SDLC) to ensure early defect detection and continuous feedback loops:

- **Sprint Planning & Requirements Analysis:** Collaborated during sprint grooming to clarify acceptance criteria and identify potential edge cases before coding began.
- **Continuous Testing (CI/CD Support):** Conducted smoke and sanity testing immediately upon new build releases to provide developers with rapid feedback.
- **Defect Lifecycle Management:** Maintained a streamlined feedback loop using Linear, ensuring every bug was documented, prioritized, and re-tested within the sprint cycle.
- **Release Gatekeeping:** Acted as the final quality gate, ensuring that features met performance, security, and UI standards before production deployment.

Defect Management Workflow

My defect management approach ensured rapid resolution and clear communication between QA and Engineering teams:

1. **Detection & Verification:** Identifying the bug and ensuring it is reproducible.
2. **Documentation:** Creating a detailed ticket in Linear including steps to reproduce, expected vs. actual results, severity level, environment details, and visual evidence (logs/screenshots/videos).
3. **Triage:** Collaborating with the product team to prioritize the defect based on impact and urgency.
4. **Fix & Resolution:** Developer implements a fix.

5. **Re-testing:** QA verifies the fix in the development/staging environment to confirm the issue is resolved and no regressions were introduced.
6. **Closure:** Updating the ticket status to "Closed" or "Verified" and updating the testing dashboard.

4. Professional Workflow

Each ticket underwent a rigorous 10-step verification process to ensure zero-defect leakage to production:

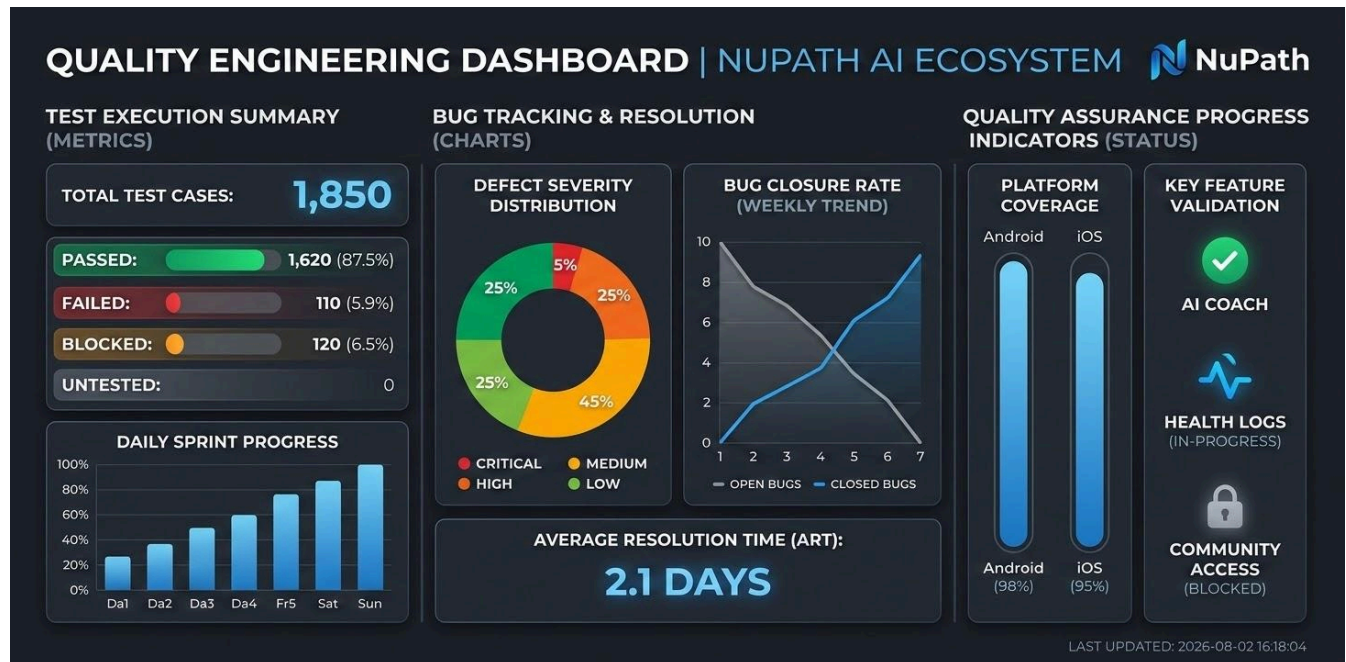
1. **Requirement Synthesis:** Reviewing tickets and acceptance criteria.
2. **Visual Audit:** Studying Figma designs for UI/UX alignment.
3. **Scenario Planning:** Drafting test cases including positive and edge-case scenarios.
4. **Execution:** Performing deep-dive functional testing.
5. **Side-Effect Analysis:** Running regression tests on impacted modules.
6. **Evidence Collection:** Recording logs, screenshots, and videos.
7. **Defect Reporting:** Creating clear, reproducible bug reports in Linear.
8. **Resolution Verification:** Re-testing issues once fixed by developers.
9. **Data Update:** Finalizing verification results and documentation.
10. **Release Readiness:** Marking tickets as "Verified" for the production pipeline.

5. Challenges & Problem Solving

During the project, I addressed several critical technical challenges:

- **AI Dynamism:** Validating unpredictable, AI-generated responses to ensure they adhered to safety and logic parameters.
- **Synchronization:** Resolving data discrepancies between wearable devices, mobile apps, and the backend.
- **Device Fragmentation:** Identifying and reproducing intermittent UI issues unique to specific Android/iOS versions.
- **Environment Stability:** Managing QA cycles within a fast-paced environment involving frequent build updates.

6. Business Impact & Outcomes



My involvement in the NuPath AI project led to tangible improvements in product quality and team efficiency:

- **Enhanced Stability:** Identified critical functional and workflow issues prior to public release, preventing negative user feedback.
- **Design Fidelity:** Guaranteed a high-quality UI/UX experience by meticulously comparing builds against Figma designs.
- **Developer Support:** Facilitated faster fix cycles by providing comprehensive technical evidence and reproduction steps.
- **Reliable Releases:** Contributed to a stable, predictable release cycle for both Android and iOS markets.

Declaration

I hereby certify that the contributions mentioned above accurately reflect my professional involvement as a QA Engineer for the NuPath AI project.

Shawon Ahmed
Software Quality Assurance Engineer